

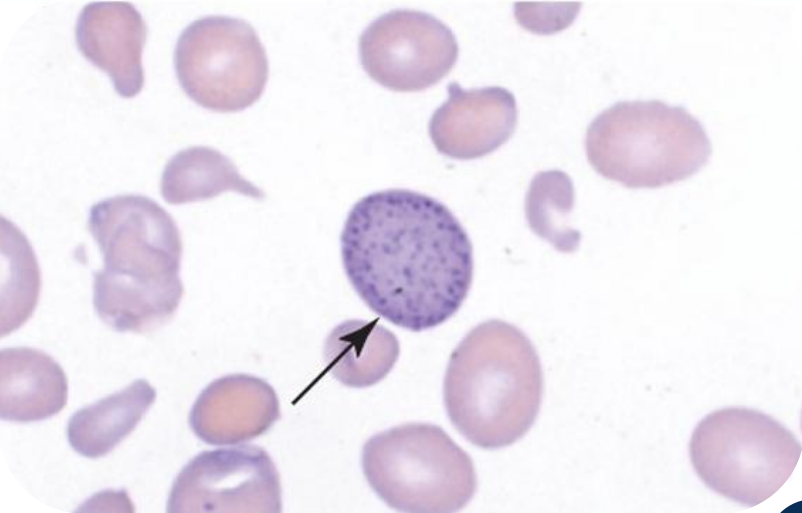
Red Blood Cell Inclusions

- RBCs do not normally contain any particulate inclusions.
 - Associated with certain disease states when present
 - Their presence may help direct further anemia investigation.
- Basophilic stippling
- Cabot Rings
- Howell-Jolly Bodies
- Pappenheimer Bodies
- Heinz Bodies

Basophilic stippling

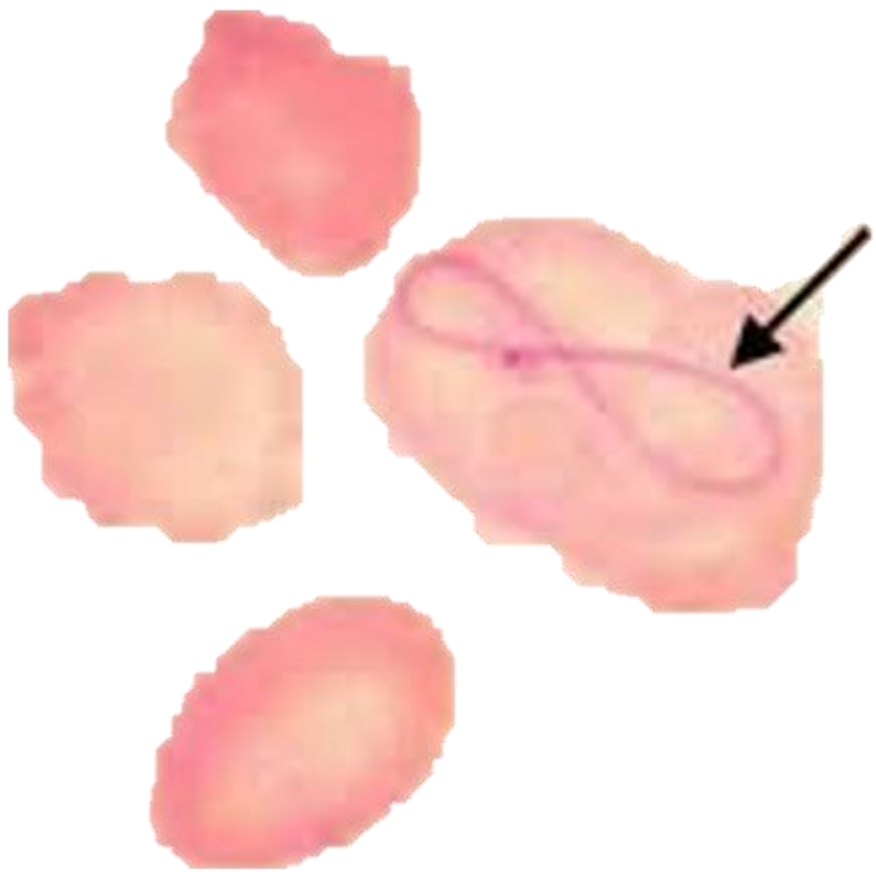
- Bluish-black granules distributed across the entire cell area
- Granules may vary in size and distribution
 - Small diffuse to coarse and punctate
 - Composed of aggregated ribosomes
- Artifact when slide is dried slowly or stained rapidly



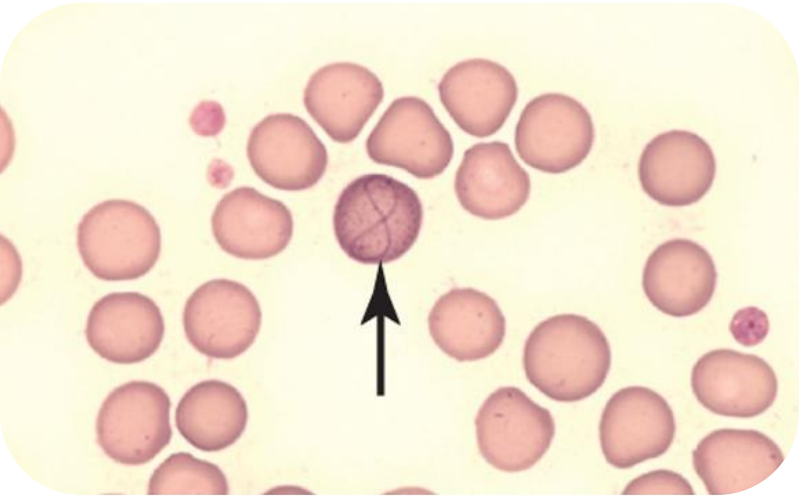
Terminology	Description	Associated Disease States	Image
Basophilic stippling	Round or irregularly shaped granules of variable number and size, distributed throughout the RBC Composed of aggregates of ribosomes (RNA) Stain bluish black with Wright's stain	Lead poisoning; anemias associated with abnormal hemoglobin synthesis; thalassemia	 A micrograph showing several red blood cells. One cell in the center is highlighted with a black arrow, showing a dense collection of small, dark, punctate granules (basophilic stippling) within its cytoplasm.

Cabot rings

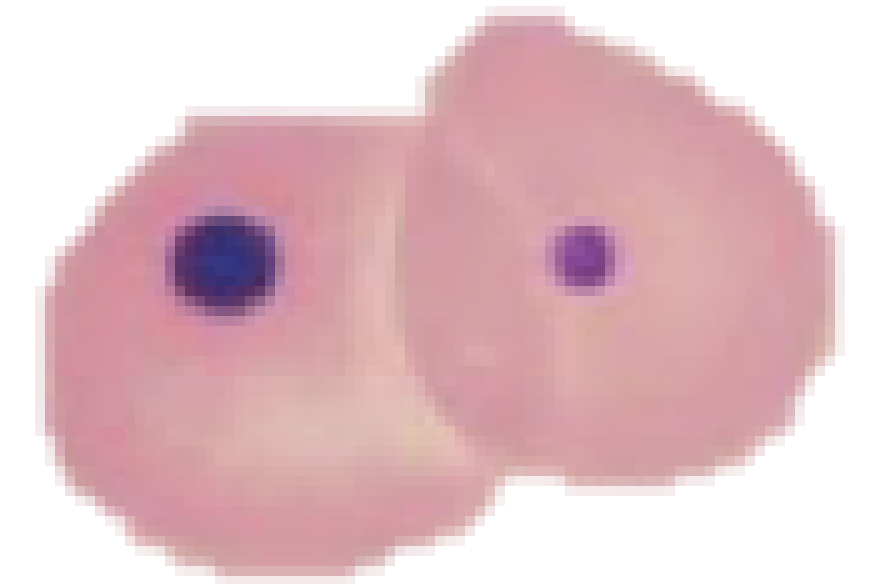
- Reddish violet figure 8 or oval ring
- Thought to be remnants of spindle fibers from mitosis
- Seen in severe anemias and dyserythropoiesis



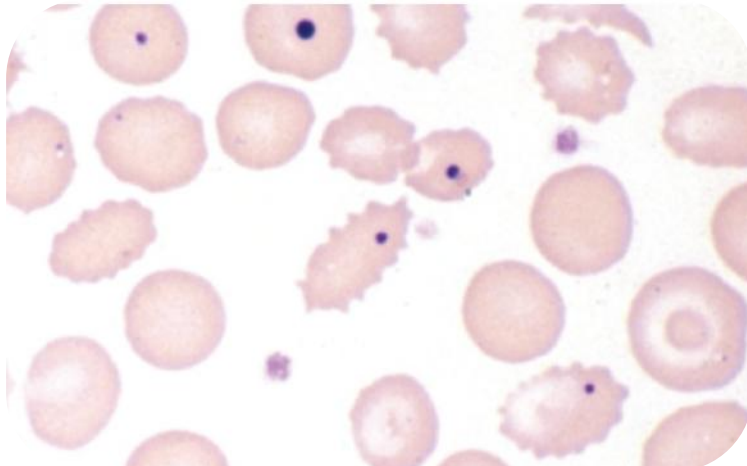
Terminology	Description	Associated Disease States	Image
Cabot rings	Appear as a figure-8, ring, or incomplete ring Thought to be composed of the microtubules of the mitotic spindle Stain reddish violet with Wright's stain	Severe anemias; dyserythropoiesis	



Howell-Jolly Bodies



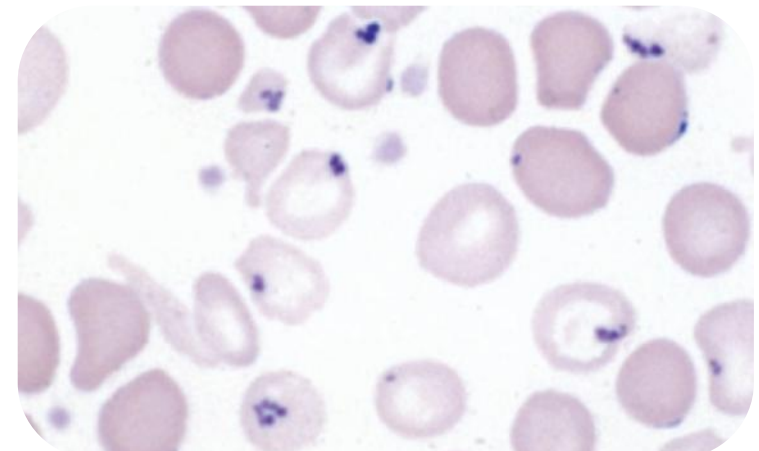
- Dark purple spherical granules
 - Nuclear (DNA) fragments
- Usually occur singly in RBCs, rarely more than two per cell
- Associated with nuclear maturation abnormalities

Terminology	Description	Associated Disease States	Image
Howell-Jolly bodies	Small, round bodies composed of DNA usually located eccentrically in the red cell Usually occurs singly, rarely more than two per cell Stains dark purple with Wright's stain	Post splenectomy; megaloblastic anemias; some hemolytic anemias; functional asplenia; severe anemia	

Pappenheimer bodies

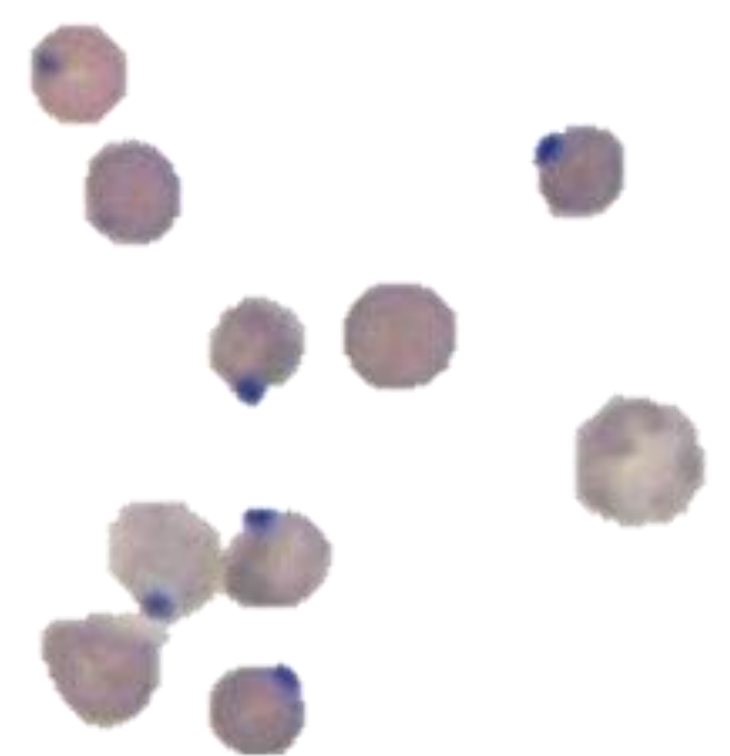
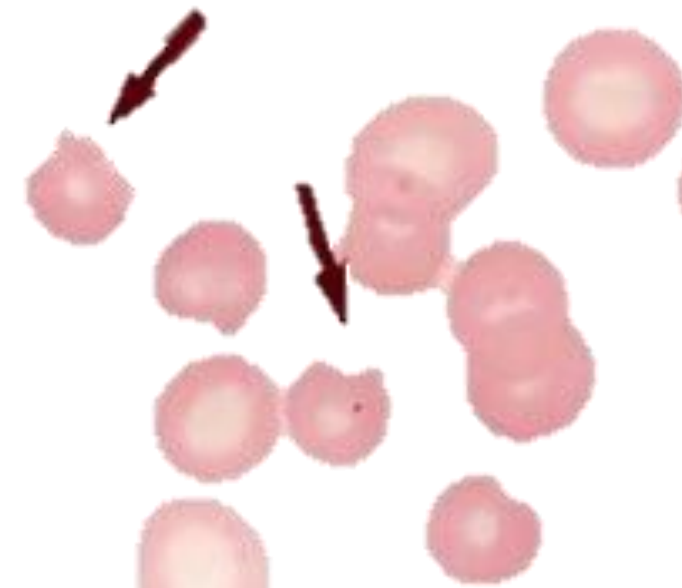


- Damaged secondary lysosomes and mitochondria
 - Variable in protein and iron composition
- Small granules (dumbbells) in RBC cytoplasm
- Romanowsky/Wright stains protein matrix of granules

Terminology	Description	Associated Disease States	Image
Pappenheimer bodies	Clusters of granules containing iron that are usually found at the periphery of the cell Visible with Prussian blue stain and Wright's stain	Sideroblastic anemia; thalassemia; other severe anemias	

Heinz Bodies

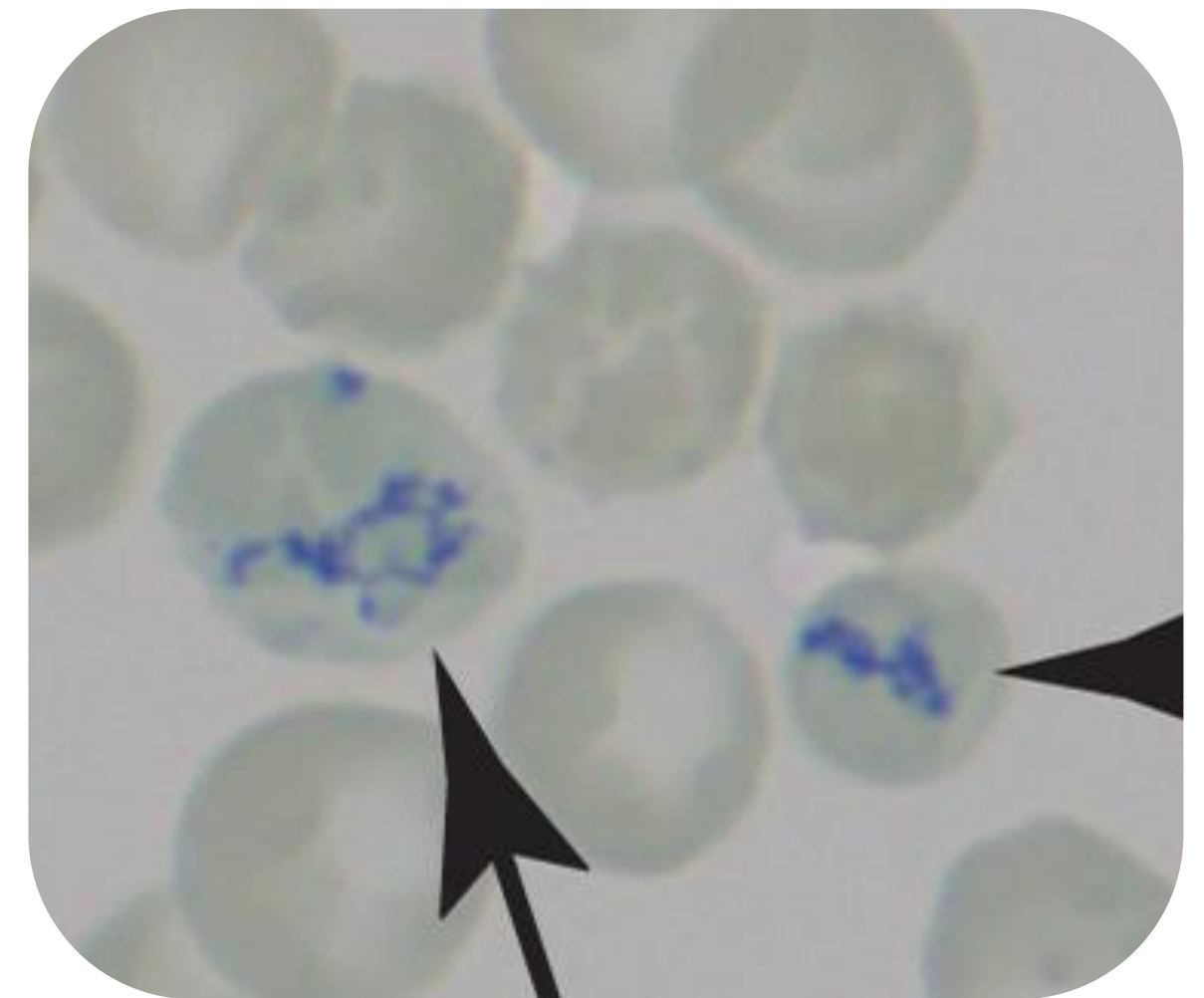
- Do not stain with Romanowsky/Wright stain
- Only stains with supravital stain
 - 2–3 μm round masses of aggregated denatured HGB
 - Under/attached to the cell membrane



Terminology	Description	Associated Disease States	Image
Heinz bodies	Bodies composed of denatured or precipitated hemoglobin Not visible on Wright's stained blood smears With supravital stain appear as purple, round-shaped bodies of varying size, usually close to the cell membrane Can also be observed with phase microscopy on wet preparations	G6PD deficiency; unstable hemoglobin disorders; oxidizing drugs or toxins; post splenectomy	

Reticulocyte

- Supravital stain precipitates ribosomes
 - Blue-black granules
 - Two or more aggregates considered when counting



Terminology	Description	Image
Reticulofilamentous substance (reticulocyte)	Artifactual aggregation of ribosomes Not visible on Wright's stained smears; supravital stain (e.g. new methylene blue) must be used Appears as deep blue reticular network	A microscopic image showing a field of reticulocytes. Several cells exhibit a deep blue, reticular network of precipitated ribosomes within their cytoplasm, characteristic of supravital staining.

Sideroblasts and Siderocytes

- Sideroblasts (nucleated RBCs)
 - $>1/3$ around perimeter
- Siderocytes (non-nucleated mature RBCs)
- Perl's Prussian blue stains iron aggregates blue

Terminology	Description
Sideroblast	Iron granules found in erythroblasts Stains blue with Perl's Prussian blue stain but not with Romanowsky stains
Siderocyte	Iron granules found in erythrocytes Stains blue with Perl's Prussian blue stain not stain with Romanowsky stains

